

1.Distinguish between synchronous and statistical TDM.

Parameter	Synchronous TDM	Statistical TDM
Working	Each input connection is assigned a fixed time slot.	Slots are allotted dynamically only to inputs that have data to send.
No. of Slots	Number of slots = number of input lines.	Number of slots is usually less than the number of input lines.
Buffering	No buffering is required.	Buffering is required to store data before transmission.
Addressing	No address is needed because the position of the slot identifies the source.	Each slot contains an address to identify the source.
Synchronization	Synchronization bits are used at the beginning of each frame.	No synchronization bits are required.
Bandwidth	Bandwidth may be wasted when an input has no data.	Bandwidth is used more efficiently because empty slots are avoided.
Data Separation	The demultiplexer extracts data from the slots in a fixed order.	The demultiplexer checks the address and sends the data to the appropriate destination.

Synchronous TDM Diagram

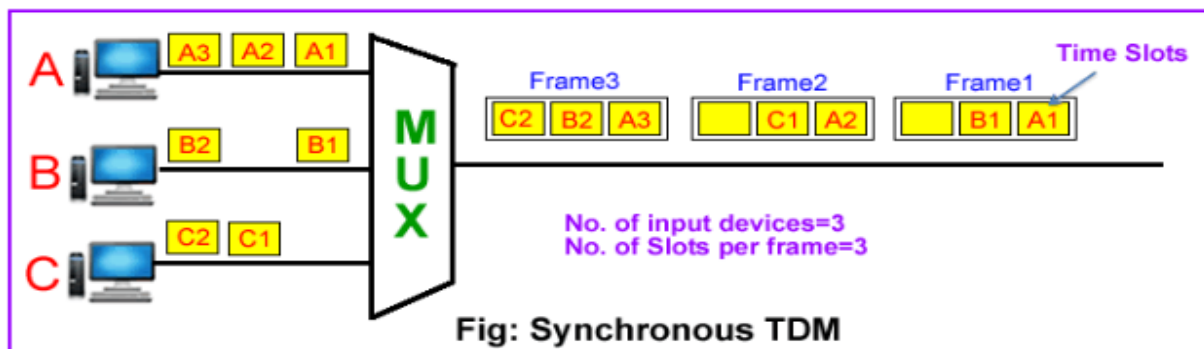


Fig: Synchronous TDM

Statistical Diagram:

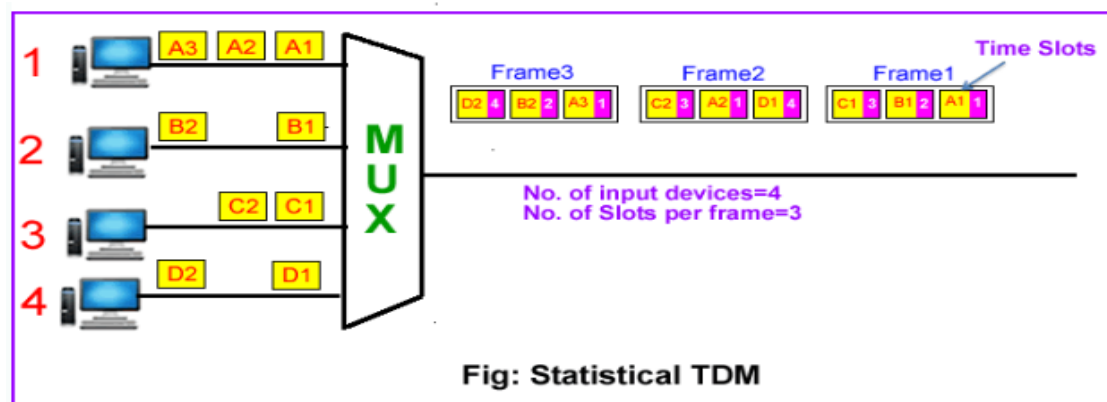


Fig: Statistical TDM

2. Explain how FHSS achieves bandwidth spreading.

FHSS (Frequency Hopping Spread Spectrum) is a spread-spectrum technique in which the source signal is transmitted by changing from one carrier frequency to another.

Bandwidth spreading process:

- The source generates the original signal.
- A pseudorandom code generator generates a k-bit pattern for each hopping period.
- The pattern selects a carrier frequency from the frequency table.
- A frequency synthesizer generates the selected carrier frequency.
- The source signal modulates this carrier and is transmitted.
- In the next hopping period, a new pseudorandom pattern selects another carrier frequency.
- Thus, the signal uses M different frequencies over time.
- Therefore, the signal is spread over a bandwidth much larger than the original signal bandwidth: $B_{FHSS} > B$

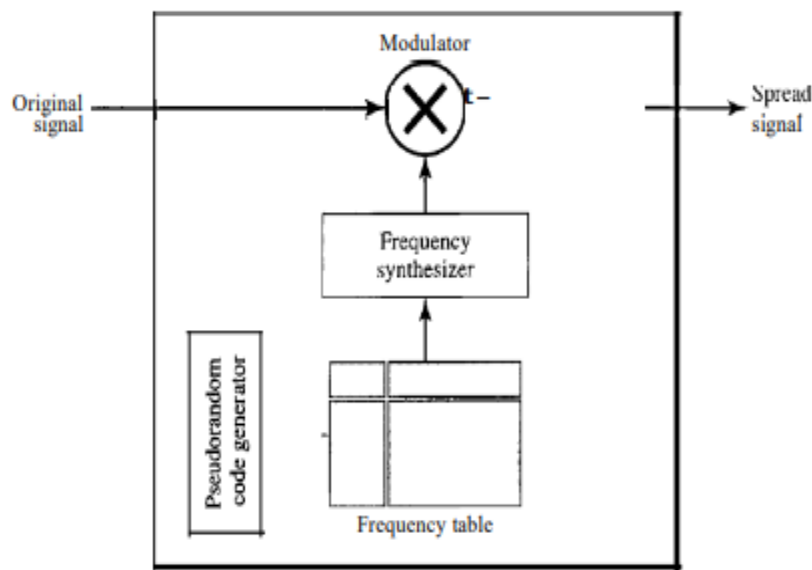


Diagram: Frequency hopping spread spectrum (FHSS)